Canvas P12 — Composition: Everything Is a Widget

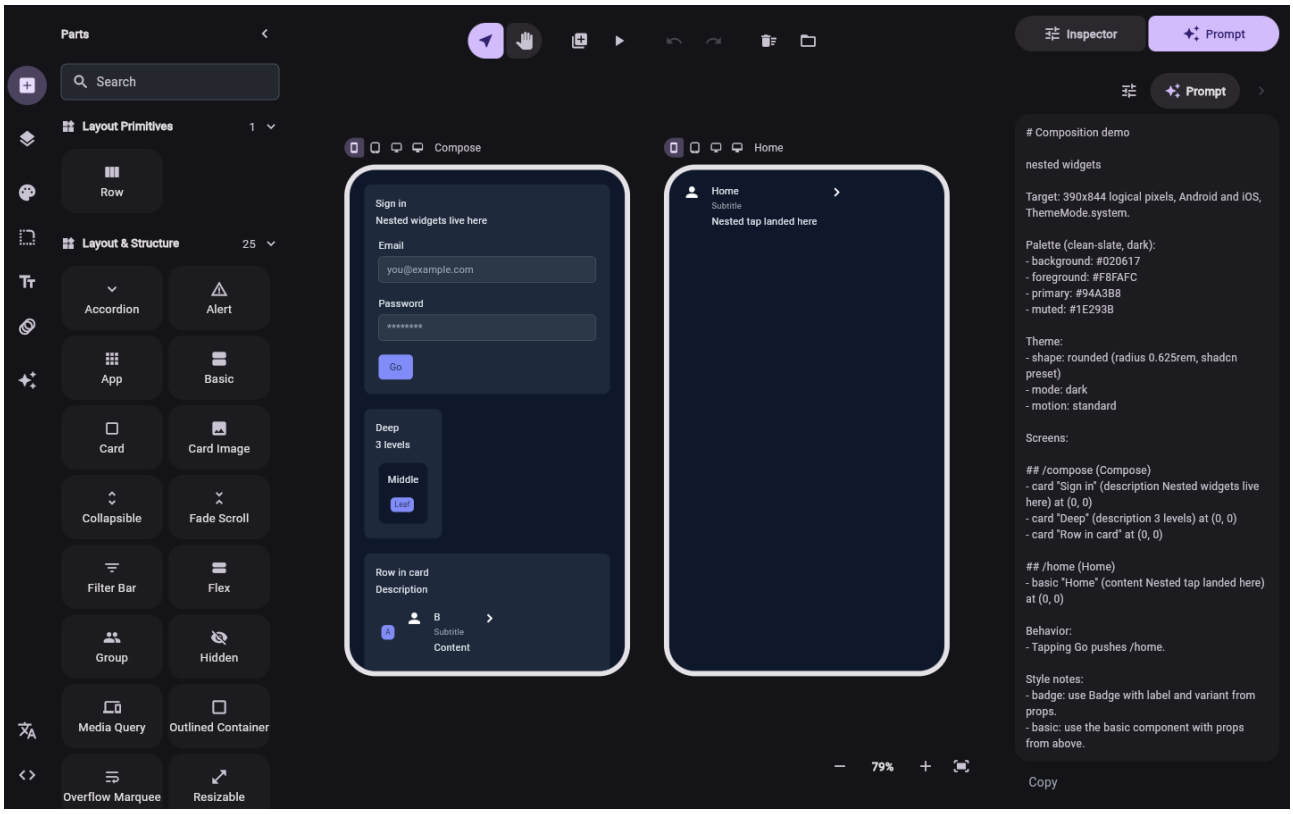
Branch feat/canvas-p3 · Flutter 3.47.2 · Release bundle, 1440×900 · Nested demo: card{inputs+button}, card>outlined>badge (3-deep), card>row{badge+basic}, tap Go→Home · Nested export: 0 errors, 0 warnings, pushNamed present

Track	What landed	Tests
C0 — spec+core	COMPOSITION.md; parentId generalizes (no schema change); children/descendants/isAncestorOf/validateNes	+26
C1 — render	NodeBuilder typedef; 10 container slots; recursive phone+preview surfaces; leaf fallback; deepest-wins selection	+12
C3 — codegen	Recursive slot splice single+bundle; per-level tiers/taps; cycle-safe; children-less byte-identical	+20
C2 — interact	reparentNode + guards; deepest-container drops; breadcrumb + parent-picker; nested layers	+30

Repo health: analyze clean both packages; app 620 pass, core 185 pass. Notable: C1 proved 1-arg builders are not subtypes of the 2-optional typedef (59 analyzer errors) — resolved via is-NodeBuilder check, palette untouched; C2 deviated deliberately (illegal reparent targets visible-but-disabled so refusals are discoverable).

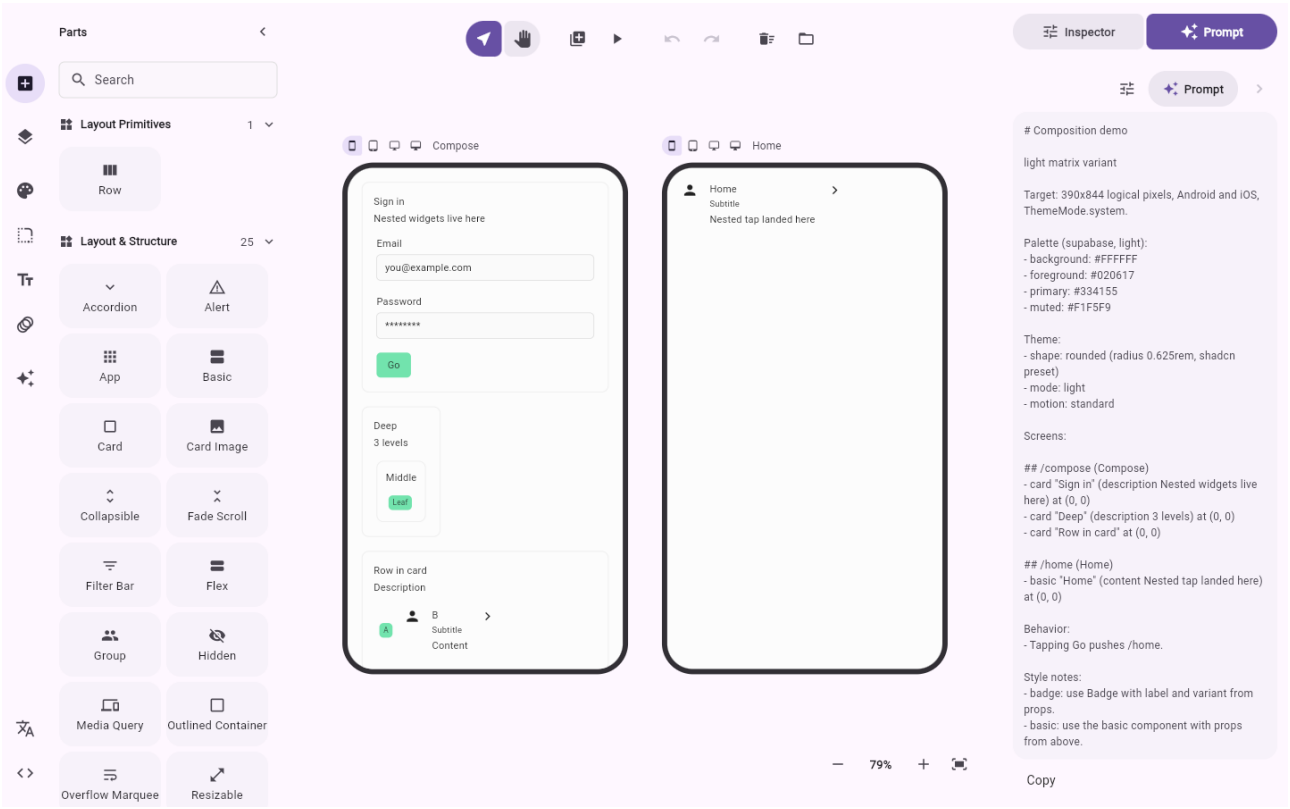
1 · Dark — nesting live (card{Email, Password, Go}; Deep>Middle>Leaf; row-in-card)

Inputs and the Go button render INSIDE the Sign-in card chrome; 3-deep chain nests visibly; avatar+basic sit side by side inside Row-in-card. Prompt panel already narrates the nested tap ("Tapping Go pushes /home").



2 · Light / supabase — same nesting, consistent

Green preset-following Go button, clean cards, no stray colors. Nested export of this doc: zero issues; navigation wired through nesting.



What this unlocks + next

Any widget can now parent any widget: cards hold forms, dialogs hold lists, rows hold cards. Next loop candidates: container promotion rule for the 58 leaf kinds (per-kind slot proof), drag-preview rendering inside target chrome, multi-select + group-into-container, nested export golden for the demo doc. Loop rule stands.